

CRICKSTATS

Cricket Player Performance Tracking Application

Salesforce

B.Sc. Computer Science Project

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PROJECT DETAILS

Project Title	Cricket Player Performance Tracking Application – CRICKSTATS
Platform	Salesforce
Frontend	Salesforce Lightning
Backend	Salesforce Objects
Database	Salesforce Cloud
Automation	Salesforce Flows
Analytics	Reports and Dashboards

ABSTRACT

CRICKSTATS is a Salesforce-based application used to store and track cricket player performance. It records player details and match-by-match performance in a structured way. The system uses custom objects, relationships, roll-up summary fields, flows, reports, and dashboards.

The main aim is to reduce manual work and make cricket statistics easier to store, search, compare, and analyze. Coaches, managers, analysts, and Salesforce administrators can use the application to view player performance and prepare reports.

The project uses Salesforce Lightning for the user interface, Salesforce Objects for data storage, Salesforce Cloud as the database, Salesforce Flows for automation, and Reports and Dashboards for analysis. The source document describes this solution and its development steps.

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1.INTRODUCTION

1.1 Project Overview

CRICKSTATS is a Salesforce application for tracking cricket player performance match by match. It stores player information, match performance, and calculated statistics. It also provides reports and dashboards so users can understand player performance easily.

1.2 Purpose

- Digitize player performance records.
- Keep cricket statistics in one place.
- Reduce errors caused by manual records.
- Provide quick reports and analytics.
- Support player development and team planning.
- Automate repeated data management tasks.

1.3 Main Users

- Team Managers
- Coaches
- Sports Analysts
- Salesforce Administrators

1.4 User Story

As a Salesforce Admin, I want to create a complete player performance tracking application in Salesforce so that match-by-match player information can be stored, analyzed, and reported easily. The application uses Player and Match Performance objects with relationships between them.

2. IDEATION PHASE

2.1 Problem Statement

In traditional cricket teams, player statistics may be stored in different files or maintained manually. This can make it difficult to get correct statistics quickly.

- Coaches may find it difficult to track performance across many matches.
- Managers may spend too much time updating player records.
- Analysts may find it difficult to compare player trends.
- Scattered data can cause errors and delays.
- A centralized system is needed for simple analysis.

2.2 Empathy Map

- Think and Feel: Users need accurate data and better insights.
- Hear: Users receive feedback from management.
- See: Users see data from many sources.
- Say and Do: Users often collect statistics manually.

- Pain: Data errors and delays.
- Gain: Faster decisions and automation.

2.3 Brainstorming Ideas

- Web-based player management system
- Mobile statistics tracker
- Salesforce-integrated CRM
- AI-based performance prediction
- Dashboard-based analytics

Selected idea: Salesforce-based performance tracking platform called CRICKSTATS.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey

1. Login
2. Enter Player Data
3. Record Match Performance
4. Generate Reports
5. Analyze Results
6. Make Decisions

3.2 Functional Requirements

Requirement	Simple Description
FR-1 Player Registration	Add and store player details.
FR-2 Match Performance Entry	Enter performance for each match.
FR-3 Dashboard Generation	Show important statistics visually.

FR-4 Report Creation	Create player and match reports.
FR-5 Flow Automation	Automate repeated tasks and updates.

3.3 Non-Functional Requirements

- Usability – The application should be simple to use.
- Security – Player data should be protected.
- Reliability – Data should be stored correctly.
- Performance – Pages and reports should respond quickly.
- Availability – The cloud system should be accessible when needed.
- Scalability – The system should support more players, teams, and records.

These requirements are based on the source document's usability, security, reliability, performance, availability, and scalability sections.

3.4 Data Flow

User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard

3.5 Technology Stack

- Frontend: Salesforce Lightning
- Analytics: Reports and Dashboards
- Automation: Salesforce Flows
- Backend: Salesforce Objects
- Database: Salesforce Cloud

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The old method depends on manual records. This can take time and may create errors. CRICKSTATS stores information in Salesforce Cloud and provides structured reports and dashboards. This makes the information easier to access and analyze.

4.2 Proposed Solution

- Player custom object
- Match Performance custom object
- Master-detail relationship
- Lookup relationship
- Roll-up summary fields
- Lightning application
- Automated flows
- Reports and dashboards

4.3 Solution Architecture

The system has four simple layers:

- Presentation Layer – Salesforce Lightning UI.
- Application Layer – Flows and business logic.
- Data Layer – Player and Match Performance objects.
- Analytics Layer – Reports and dashboards.

4.4 Main Data Relationship

Player is the parent record and Match Performance stores the related match information. The source specifies a Master-Detail relationship from Match Performance to Player and a Lookup relationship involving Match Performance.

5. PROJECT PLANNING AND SCHEDULING

5.1 Development Method

The project follows an Agile-based approach. Development is divided into smaller tasks and milestones.

5.2 Main Milestones

7. Create Salesforce Developer Account
8. Create Player Object
9. Create Match Performance Object
10. Create Custom Tabs
11. Create CRICKSTATS Lightning App
12. Create Fields and Relationships
13. Create Roll-up Summary Fields
14. Create Flows
15. Create Reports and Dashboards
16. Test the Application

5.3 Sprint Plan

Sprint	Main Work	Goal
Sprint 1	Developer setup and data modeling	Create Salesforce account and objects
Sprint 2	Navigation and UI	Create tabs and Lightning app
Sprint 2	Data structure	Create fields and page layouts
Sprint 3	Automation	Create validation and flows
Sprint 4	Testing and reporting	Test system and prepare reports

6. PROJECT DEVELOPMENT AND IMPLEMENTATION

6.1 Salesforce Developer Account

- Open the Salesforce Developer signup page.
- Enter name and email details.
- Select a developer role.
- Enter organization details.
- Create a Salesforce username.
- Activate the account from the email.
- Set a password and security question.
- Open Salesforce Setup.

6.2 Player Object

Create a custom object named Player. Use the plural label Players. Create Player Name as the record name. Enable useful options such as reports, activities, field history, and search.

6.3 Match Performance Object

Create a custom object named Match Performance with the plural label Match Performances. This object stores the performance information for individual matches.

6.4 Custom Tabs

Create custom tabs for Player and Match Performance. Add the tabs so users can easily open and manage records.

6.5 CRICKSTATS Lightning App

- App Name: CRICKSTATS
- Navigation: Standard navigation
- Form factors: Desktop and phone
- Navigation items: Players, Match Performances, Reports, Dashboards
- User profile: System Administrator

The source document gives these Lightning App setup values.

6.6 PLAYER FIELDS

Field	Type	Purpose
Player Name	Text	Stores player name
Player Number	Text	Stores player number
Profile Picture	Rich Text Area	Stores profile picture information
Date of Birth	Date	Stores date of birth
Role	Picklist	Batsman, Bowler, Wicket Keeper, All Rounder
Batting Style	Picklist	Right-Handed or Left-Handed
Bowling Style	Picklist	Different bowling styles
Nationality	Picklist	Stores player's country

Team	Picklist	Stores team name
AVG Score	Formula	Calculates average score
Match Performance	Lookup	Links player-related performance

6.7 MATCH PERFORMANCE FIELDS

Field	Type
Match Performance Name	Text
Player of the Match	Checkbox
Match Result	Picklist
Runs Scored	Number
Balls Faced	Number
Fours	Number
Sixes	Number
Overs Bowled	Number
Wickets Taken	Number
Runs Conceded	Number
Maiden Overs	Number
Stumpings	Number
Catches	Number
Player	Master-Detail

These fields are simplified from the field lists in the source document.

6.8 ROLL-UP SUMMARY FIELDS

- Matches Played – Count of Match Performance records.
- Number of 50's – Count of records meeting the 50-run condition.
- Number of 100's – Count of records meeting the 100-run condition.
- Total 4's – Sum of Fours.
- Total 6's – Sum of Sixes.
- Total Runs – Sum of Runs Scored.
- Total Overs – Sum of Overs Bowled.
- Total Wickets – Sum of Wickets Taken.
- Total Stumpings – Sum of Stumpings.
- Total Catches – Sum of Catches.

6.9 AVG SCORE FORMULA

The source document gives the following formula:

```
IF(
   Matches_Played__c > 0,
   Total_Runs__c / Matches_Played__c,
   0
)
```

Important: Salesforce must use the actual API names generated in your org. If Matches_Played__c does not exist, use Insert Field in the Formula Editor to insert the real Matches Played field. The source document specifies the formula and the roll-up fields, but the exact API names may differ if the fields were created differently in your Salesforce org.

6.10 FLOW AUTOMATION

Player Flow

→ Open Setup and search for Flow.

→ Create a new Screen Flow.

→ Add a screen component.

→ Add a Team picklist.

→ Create a Picklist Choice Set from the Player Team field.

→ Use Get Records to find players for the selected team.

→ Show the players in a Data Table.

→ Display useful columns such as Player Name, Total Runs, AVG Score, Batting Style, Bowling Style, Matches Played, 50s, and 100s.

→ Save the flow as Player Flow.

→ Activate the flow.

The source document describes a Screen Flow that filters Player records by Team and displays selected player information in a data table.

6.11 Home Page Flow

- ❖Open the CRICKSTATS navigation bar.
- ❖Add the Home page.
- ❖Remove unnecessary components.
- ❖Add the Flow component.
- ❖Select the Players Flow.
- ❖Save and activate.
- ❖Assign it to the CRICKSTATS app.

7. FUNCTIONAL AND PERFORMANCE TESTING

7.1 Functional Testing

Test	Expected Result	Status
Create Player	Player record is saved	Pass
Create Match Performance	Match record is saved	Pass
Enter required data	Validation works correctly	Pass
Calculate total runs	Roll-up shows correct total	Pass
Calculate matches played	Count is updated	Pass
Open Player Flow	Flow displays players	Pass

Filter by Team	Only selected team players appear	Pass
View report/dashboard	Statistics are displayed	Pass

7.2 Performance Testing

- Check page loading speed.
- Check report generation speed.
- Check dashboard loading.
- Test multiple user actions.
- Check data consistency.
- Verify that the application works without unnecessary delays.

The source document states that testing covered record creation, validation, record types, flows, roll-up calculations, report generation, dashboard loading, and system response.

8. ADVANTAGES AND DISADVANTAGES

8.1 Advantages

- Centralized system – all important data is stored in one place.
- Real-time analytics – reports and dashboards help users see current information.
- Secure platform – Salesforce provides access controls.
- Scalable design – the system can grow with more records and users.
- Reduced manual work – automation reduces repeated tasks.

8.2 Disadvantages

- Requires a Salesforce license or suitable Salesforce environment.
- Internet connection is needed for cloud access.
- Offline access is limited.

These advantages and disadvantages are listed in the source document.

9. CONCLUSION

CRICKSTATS is a Salesforce-based cricket player performance tracking application. It stores player and match information in a structured way. Custom objects, relationships, roll-up summaries, flows, reports, and dashboards help automate the system.

The application reduces manual work and makes player statistics easier to access. It also demonstrates practical use of Salesforce, SDLC, Agile development, CRM concepts, cloud storage, and reporting.

The source document concludes that the system improves data management and reporting accuracy and supports better decision-making.

10. FUTURE SCOPE

1. Mobile Application Integration

Users can access player statistics through a mobile application.

2. AI-Based Performance Prediction

AI can be used to predict future player performance.

3. Video Analysis

Match videos can be connected with player statistics for better analysis.

4. Player Ranking System

Players can be ranked based on their performance.

5. Public Statistics Portal

Selected player statistics can be displayed on a public website.

6. Multi-League Support

The system can support multiple cricket leagues and teams.

7. Advanced Dashboards

More charts and visual reports can be added.

8. Performance Comparison

Users can compare two or more players using statistics.

These points follow the future-scope section in the CRICKSTATS project document, which lists mobile integration, AI predictionvideo analysis, player ranking, a public statistics portal, and multi-league support.

11. APPENDIX

The appendix contains supporting information and project materials.

11.1 Salesforce Objects

- Player Object – Stores player details.
- Match Performance Object – Stores match-by-match performance details.

11.2 Important Player Fields

- Player Name
- Player Number
- Date of Birth
- Role
- Batting Style
- Bowling Style
- Nationality
- Team
- AVG Score
- Matches Played
- Total Runs

- Total 4's
- Total 6's
- Total Wickets

11.3 Match Performance Fields

- Match Performance Name
- Player of the Match
- Match Result
- Runs Scored
- Balls Faced
- Fours
- Sixes
- Overs Bowled
- Wickets Taken
- Runs Conceded
- Maiden Overs
- Stumpings
- Catches
- Player

11.4 Flow Configuration

The Player Flow allows the user to select a team and display the players belonging to that team. It can show information such as player name, total runs, average score, batting style, bowling style, matches played, 50s, and 100s.

11.5 Project Screenshots

1. Salesforce Setup

2. Player Object

3. Match Performance Object

4. Player Fields

5. Match Performance Fields

6. Relationships

7. Roll-up Summary Fields

8. Formula Field

9. CRICKSTATS Lightning App

10. Player Flow

11. Reports

12. Dashboard

11.6 Sample Data

Add sample Player and Match Performance records to demonstrate the working of the application. Records should be entered in both objects to obtain results.